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Tight Error Bars, Wrong Parts

A fit to photographs cannot tell you when it found the wrong CAD program

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Abstract

We recover editable CAD programs (a plate with holes, counterbores, slots, bosses and patterns, each with its dimensions) from photographs by analysis by synthesis: a search over programs scored by description length, with every candidate's dimensions fitted through a differentiable renderer. Such a fit reports a Fisher error bar for every dimension. We ask whether those error bars, or the fit's residuals, tell a right answer from a wrong one. On held-out synthetic parts they do not. Among 76 parts, 17 of the 37 whose every error bar is under 1 mm are wrong, and the largest error bar separates wrong answers from right ones with an area under the ROC curve of 0.58. Four image-based checks (residual on the part against the table, outline offset, unexplained residual blobs, score margin to the runner-up) do little better: with thresholds fitted on half of 100 fresh parts, they flag 4 of 37 wrong answers on the other half. Most wrong answers explain the photographs. The one reliable signal is outside the fit: the number of photographs. We report the recovery's accuracy plainly (41 of 100 fresh parts within 1 mm on every dimension) and a continuous relaxation over features that made identification worse. Every photograph is synthetic.

1 Introduction

A drawing of a part is a short program: “a 69 by 56 by 7 mm plate, chamfered, with a 2.6 mm hole and a boss” is a dozen numbers. Recovering that program from photographs takes two steps. **Identification** decides which features the part has. **Measurement** decides their dimensions. Analysis by synthesis does both by rendering candidate programs and keeping the one whose render best explains the photographs[1,2]. With a differentiable renderer[3] the measurement step is a least-squares fit, and least squares gives every dimension an error bar from the Fisher information.

A user who gets back a program with every dimension at ±0.1 mm will trust it. This paper is about why they should not. The Fisher error bar is the curvature of the score at the answer the search found. It says how well that answer is pinned if the model is right, and nothing about whether the right features were found[4]. A plate 2 mm too short with its hole missed can sit in a basin as narrow as the true one.

We measure how often this happens and whether anything computed from the fit can catch it. Our contributions are:

- **A recovery pipeline** that searches CAD programs by description length[5] through a differentiable renderer, as a testbed (Section 2).
- **A held-out measurement of when its error bars mislead.** On fresh synthetic parts, tight error bars are nearly uninformative about whether the answer is right (Section 4).
- **A negative result for post-hoc checks.** Four residual-based checks, with thresholds fitted on one half of fresh parts and tested on the other, catch few wrong answers. Part 3000 shows why: its wrong answer leaves a clean residual (Section 5).
- **What does help.** More photographs. One photograph per part is far worse than three, and a photo-count rule is the only check we keep (Section 6).

Every photograph here is rendered, so every result excludes what a real camera, print and room would add. The claim is about the fit’s self-assessment, which synthetic data tests cleanly, because the truth is known exactly.

2 The recovery

A program is a plate (width, depth, height, and a sharp, filleted or chamfered corner) with an ordered list of features on its top face (through-holes, counterbores, slots, bosses, linear and circular hole patterns) and its pose on the table. We score a program in nats,

$$\text{score} = \underbrace{-\log p(\text{photos} \mid \text{render}(\text{program}, \theta))}_{\text{residual}} + \underbrace{\frac{1}{2}\log\det F(\theta)}_{\text{parameters}} + \underbrace{-\log p(\text{program})}_{\text{program}},$$

with a Huber noise model, a Laplace parameter cost from the Fisher information $F = J^\top \Sigma^{-1} J$[6,7], and a uniform grammar prior. The noise variance is scaled by the residual’s

autocorrelation area, measured on the open table, so smooth shading errors are not counted as independent pixels.

The search starts from a box, proposes edits (add a feature at a residual blob, remove one, turn a hole into a counterbore or slot, merge holes into a pattern, change the corner), fits each by Levenberg–Marquardt and keeps a beam of two. The best four distinct programs are refitted at full resolution and the lowest score wins. The Jacobian comes from edge sampling[3] in irys[8], through a deterministic photo model (closed-form shading, spherical-harmonic light, one key light with soft shadows, and ambient occlusion whose derivative we take by treating its 16 fixed rays as directional lights). A printed AprilTag[9] fixes each camera and the metric scale.

Test photographs are irys path-traced renders of a part in PLA of one of six colours beside the tag, under a random environment or a synthetic room, at 2016×1512, passed through a phone camera model (defocus, shot and read noise, tone curve, JPEG). Parts are drawn from the grammar's generator. Every setting was frozen on a 20-part development set before any held-out run.

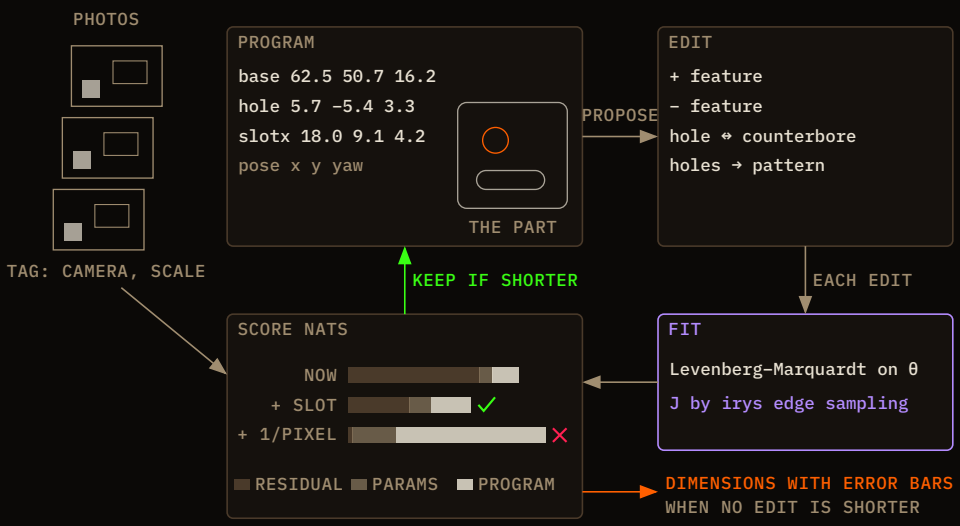


FIG 1 The recovery loop. The search edits a CAD program, fits each edit's dimensions through the renderer's edge-sampled Jacobian, scores it in nats, and keeps it only if the description of the photographs gets shorter. Bar lengths are schematic.

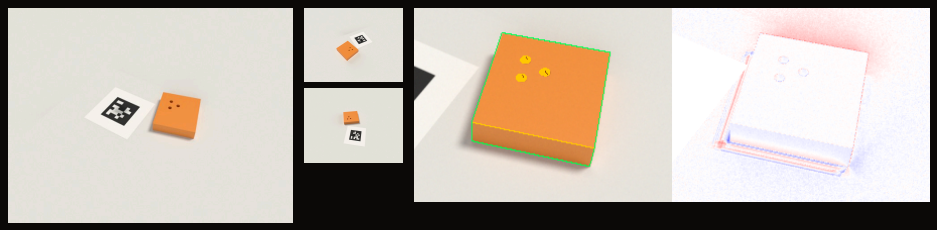
3 Accuracy

TABLE 1 Recovery accuracy on held-out synthetic parts. Worst dimension per part, after the plate's quarter-turn symmetry. Wilson 95% intervals in parentheses. Most failures are wrong features, chiefly a hole read as a counterbore or the reverse; among parts with the right features the median worst-dimension error is 0.39 mm. `data/measured.tsv`.

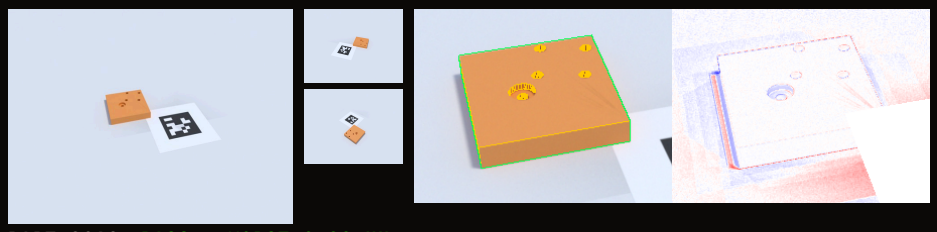
SET (3 PHOTOS EACH)	WITHIN 1 MM	RIGHT FEATURES
held-out parts, first run (80)	37 (36–57%)	44 (44–65%)
fresh parts, seeds 4000–4099 (100)	41	54

About two in five parts come back within 1 mm on every dimension, and about half with the right features (Table 1). We compare against no other method; the point of

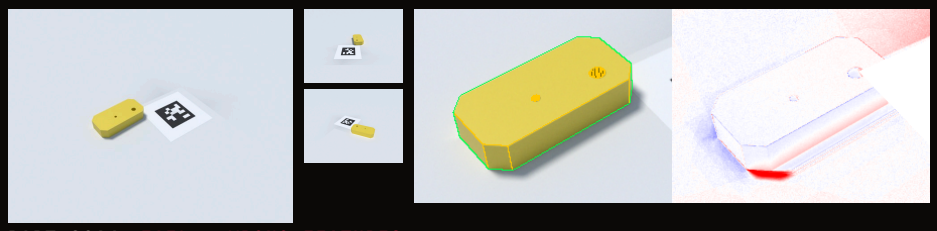
this paper is not the accuracy but what the fit says about it. Figure 2 shows two parts it gets right and one it gets wrong.



PART 3009 PASS · WORST 0.16 MM
TRUTH base 74.100 67.500 17.100; circular 3 10.400 10.300 8.600 32.200 3.000
FOUND base 73.998 67.343 16.980; circular 3 10.388 10.196 8.570 32.137 3.013



PART 3012 PASS · WORST 0.32 MM
TRUTH base 65.400 61.100 11.000; circular 4 15.900 -13.000 14.500 30.800 2.500; counterbore -1.500 7.900 2.400 5.600 3.100
FOUND base 65.348 60.992 10.774; circular 4 15.904 -12.950 14.496 120.775 2.494; counterbore -1.450 7.988 2.288 5.584 2.780



PART 3016 FAIL · WRONG FEATURES
TRUTH base 81.900 41.100 16.000 chamfer 7.800; hole -24.500 7.400 3.800; hole 7.600 3.200 2.000
FOUND base 81.490 40.885 15.903 chamfer 7.662; hole 7.560 3.272 2.027; counterbore -24.380 7.465 2.105 3.819 7.940

FIG 2 Three development parts, three synthetic photographs each. Right of the photographs, the fit's overlay: the recovered outline on the crop, and the residual (red brighter, blue darker than the model). Parts 3009 and 3012 pass at 0.16 and 0.32 mm. Part 3016 fails: its hole is read as a counterbore of the hole's outer radius, the commonest confusion. `data/recon-examples.json`.

4 Error bars do not flag wrong answers

The Fisher standard deviations are too small by a median factor of 6 and do not rank the errors (Spearman $\rho = -0.15$). One scale, $\sigma' = 17.8\sigma_F$, fitted on even seeds restores 68% coverage on odd seeds, as any fitted scale must; because σ_F does not rank errors, σ' is an average error with a unit, not a per-dimension error bar. The cause we suspect is bias from the appearance model, which moves an edge the same way in every photograph, so more pixels shrink σ_F without shrinking the error.

The worse failure is that σ is conditional on the features being right. Among the 76 held-out parts, 37 have $\sigma' \leq 1$ mm on every dimension, and 17 of those are wrong, 14 with the wrong features. As a test for a wrong answer, a part's largest σ has

an area under the ROC curve of 0.58, where 0.5 is chance. This is the gap between aleatoric and model uncertainty[10], in a setting where the model is a discrete program. Figure 3 shows every part.

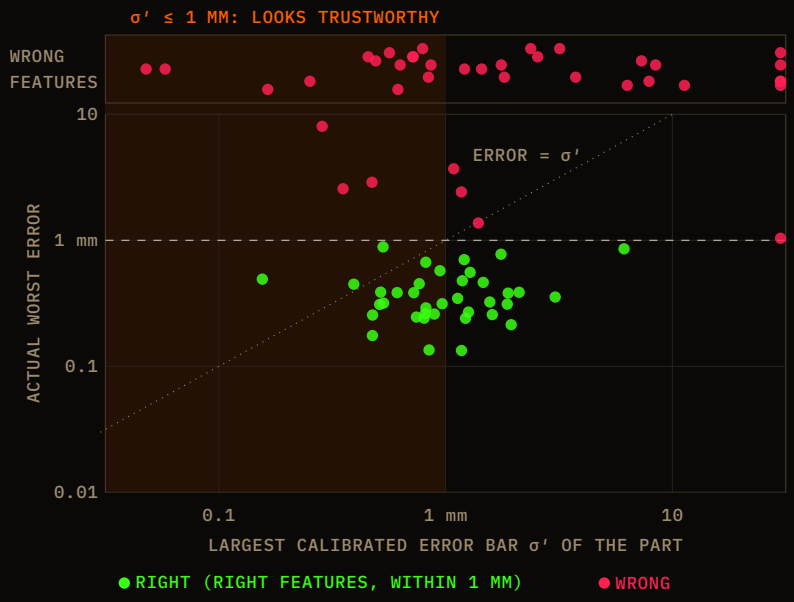


FIG 3 Error bars against errors. Each of the 76 held-out parts at its largest calibrated error bar $\sigma' = 17.8\sigma_F$ and its actual worst error; parts with the wrong features, which have no worst error, sit in the top band. In the shaded region every error bar is under 1 mm, and 17 of its 37 parts are wrong. `data/sigma-vs-error.csv`.

5 Residual checks do not either

If the error bars cannot see a wrong program, perhaps the residual can. A missing or misplaced feature should leave residual on the part that the open table does not have. We computed four signals on each finished recovery:

- **residual excess**: mean robust residual near the part over the same on the open table;
- **outline offset**: mean disagreement between the model's silhouette and the segmented part, in pixels;
- **unexplained blob**: the largest residual blob left on the top face, as an equivalent radius;
- **finalist margin**: the score gap in nats from the winner to the runner-up program.

On 100 fresh parts we fitted thresholds on the even seeds, allowing at most 20% false alarms, and tested on the odd seeds. The fitted rule caught 4 of 37 wrong answers with 0 false alarms among 13 right ones. Looser hand-set thresholds caught 14 of 37 at 3 of 13 false alarms. Neither is a usable trust signal (Figure 4). (The odd half happens to be harder, 37 wrong of 50 against 22 on the even half, so these rates are noisy.)

Part 3000 shows why. Its recovered plate is 2.15 mm too narrow and 1.08 mm too tall, and its hole is missing. Every Fisher σ is at most 0.11 mm, the residual on

the part is 0.75 times the table's (that is, no excess), and no blob remains. From its photographs' viewpoints the wrong program explains the images. No function of the residual can flag an answer that leaves no residual.



FIG 4 Residual checks on 100 fresh parts. Each dot is a part, split by whether the recovery was right. The two groups overlap on every signal; a threshold that flags many wrong parts flags many right ones too. The fourth signal, an unexplained residual blob, was zero on 97 of the 100 parts and is not drawn; its fitted threshold flags two wrong parts and no right ones. `data/check-fresh.csv`.

6 Photographs do

What separates part 3000's answer from the truth is a view the photographs did not take. On the 20-part development set, one photograph per part put 6 within 1 mm and three photographs 13; the features were right on 11 against 15. [TODO: same comparison on the fresh parts, 1 photo vs 3, run in progress on mew (runs/one).] The only check we keep in the released code flags any recovery from fewer than three photographs.

7 A relaxation that made identification worse

Since most failures are a hole-or-counterbore choice, we tried making that choice continuous. Each loose hole or counterbore became a pair of alternatives with amplitudes $a \in [0, 1]$ on a simplex; the model image blends renders without and with each, $I = I_0 + \sum_k a_k (I_k - I_0)$, whose derivative in a_k is exact; the score charges $a_k L_k$ plus an anneal $\beta a(1-a)$ that drives amplitudes to 0 or 1[11,12]. On three small problems it behaved. On the 80 held-out parts, paired against the greedy choice on the same recoveries, it found the right features on 37 against 44, fixing 0 and breaking 7; on the fresh parts, 50 against 54, fixing 1 and breaking 4. The main cause is the anneal: a real hole whose two alternatives split the evidence near 0.45 each loses both. Annealing presence and choice separately ($a_{\text{hole}} = pq$, $a_{\text{cbore}} = p(1-q)$) is the obvious fix and is untested.

8 Limitations

- **Synthetic photographs.** Rendered by the renderer the method is built on. Real photographs add appearance error, which we expect to make error bars less informative, not more.
- **One family of parts**, a plate with top-face features.
- **No baselines.** The accuracy in Table 1 is not compared with any other method.
- **Single runs.** The greedy search is not deterministic across machines, and we have not measured that variance.

9 Conclusion

In analysis-by-synthesis recovery of CAD programs, the fit's own error bars and residuals say little about whether the program is right. On held-out synthetic parts, tight error bars are wrong almost as often as loose ones, and residual checks fitted on one half catch few wrong answers on the other, because most wrong answers explain the photographs they were fitted to. What does help is evidence the fit has not seen: more photographs. A system that reports CAD dimensions with error bars should say plainly that those bars assume the features are right.

The code is the `irys-recon` crate of irys (Apache-2.0); each row of `data/measured.tsv` names the commit and run that produced it.

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